RMF Assessment: The State Institute of Science and Technology

Categorize Task #1:

1. Determine whether the controls for Confidentiality, Integrity, and Availability should be applied at the Low, Moderate, or High level. Please justify your CIA rating for each information type and make sure to note your constraints and assumptions as well. Use your determinations to fill out the table below:

	Criteria			**Justification**	**Assumptions/Constraints**
	Confidentiality	***Integrity***	***Availability***		
SIST AWS Applications					
HR Application					
HR Strategy	Moderate	Moderate	Low	Disclosure could reveal internal workforce plans; unauthorized changes could affect staffing decisions; temporary unavailability would have a limited operational impact.	Internal planning data only
Benefits Management	Moderate	High	Moderate	Disclosure could expose employee financial information; unauthorized modifications could cause incorrect benefit payments; outages could delay benefits processing.	Employees' financial records
Employee Performance Management (w/ PII)	High	High	Moderate	Disclosure could expose sensitive employee records; unauthorized modifications could affect employment decisions; temporary outages could delay HR operations.	Sensitive employee PII data stored
GMAT Application					

System and Network Monitoring	Moderate	High	High	Loss of integrity could conceal security incidents; prolonged unavailability could delay threat detection and response	Continuous monitoring is required
System Development	Moderate	High	Moderate	Disclosure could expose source code; unauthorized modifications could introduce vulnerabilities; outages could delay development activities.	Secure development lifecycle
Vulnerability data	High	High	High	Disclosure could help attackers identify weaknesses; modification could conceal vulnerabilities; unavailability could delay critical remediation	Sensitive vulnerability details stored
FCIN Application					
Funds Control	High	High	High	Disclosure could expose sensitive financial information; unauthorized changes could enable fraud or incorrect fund allocation; outages could interrupt critical financial operations.	Government financial operations
Payments	High	High	High	Disclosure could expose payment data; unauthorized modification could cause financial loss; prolonged outages could interrupt critical payment operations	Includes payment processing data
SIST Data Center Applications					
FORD Application					
Training & Employment	Moderate	Moderate	Low	Disclosure could expose employee training records; unauthorized modifications could affect	No classified workforce data

				employment records; temporary unavailability would have limited operational impact.	
CHEVY Application					
Manufacturing	Moderate	High	High	Disclosure could expose operational information; unauthorized changes could affect production accuracy and safety; outages could disrupt critical manufacturing operations.	Production supports mission operations
Construction	Moderate	High	Moderate	Disclosure could expose infrastructure design information; unauthorized modifications could create safety risks; outages could delay construction activities.	Infrastructure design data stored

System Scope and Assumptions

This academic assessment evaluates the fictional SIST environment. The AWS-hosted applications include HR, GMAT, and FCIN. FORD and CHEVY are assessed as SIST Data Center applications.
The system-level High impact conclusion is supported by the AWS-hosted applications' CIA ratings. The inclusion of data center assets in the broader authorization boundary should be reconciled with the System Security Plan.

2. Determine a high-water mark for each Application and then determine the Security Categorization for the SIST Web Infrastructure System. Please justify your high-water mark for each Application.

Application	**High Water Mark**	**Justification**
HR Application	High	Employee performance information has High confidentiality and integrity impact ratings because unauthorized disclosure or modification could significantly affect sensitive personnel records.
GMAT Application	High	Monitoring and vulnerability information has High integrity and availability impact ratings because compromised or unavailable security information could affect threat detection and remediation.
FCIN Application	High	Funds control and payment information has High confidentiality, integrity, and availability ratings because security incidents could significantly affect financial operations.
Ford Application	Moderate	Training and employment information has Moderate confidentiality and integrity ratings, with no High impact ratings identified.
Chevy Application	High	Manufacturing information has High integrity and availability ratings because unauthorized changes or outages could significantly affect production and safety.
SIST Web Infrastructure	The AWS-hosted applications include information types with High impact ratings, resulting in an overall High system categorization under the documented assumptions.	
System Security Categorization	Based on the high-water-mark approach, the SIST AWS Web Infrastructure System is categorized as High impact. The assessment identified High confidentiality, integrity, and availability impact ratings within the HR, GMAT, and FCIN applications. FORD and CHEVY were evaluated separately as data center applications. The final system boundary and security categorization will be documented consistently in the System Security Plan.	